

A Fairness-Aware Constraint Satisfaction Framework for University Exam Timetabling with Dual-Role Instructor Conflict Modeling

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Abstract—Exam scheduling at universities is an NP-hard problem due to high levels of constraint interaction. In this paper, we design a framework based on the Constraint Satisfaction Optimization Problem (CSOP) that addresses complex, realistic university settings including dual-role instructor conflicts, such as when a Ph.D. candidate acts as both an examinee and an invigilator. The CSOP design minimizes the maximum invigilator workload difference via a min-max proxy objective. Our model is solved by the Google OR-Tools CP-SAT solver and provides a multi-room Boolean assignment matrix for handling large-enrollment examinations that require dynamic partitioning across multiple venues. Experiments are conducted on four instances of the Carter/Toronto benchmark suite and on real data from a metropolitan university. Results show that the proposed framework satisfies all hard constraints across every tested instance, achieves near-optimal workload fairness (min-max gap 0–3 on three of four benchmarks), and is scalable to 189 examinations within practical time limits. A structural scalability boundary is identified at approximately 300 examinations for the current monolithic formulation. An ablation study demonstrates a fundamental tradeoff between consecutive invigilation avoidance and long-term fairness.

Index Terms—Constraint Satisfaction, Exam Timetabling, Multi-Objective Optimization, CP-SAT, Educational Administration, Workload Fairness, Constraint Programming

I. INTRODUCTION

University Exam Timetabling involves scheduling examinations within strictly defined timeslots and physical rooms such that no operational conflicts occur. This process is a cornerstone of modern educational administration, ensuring optimal use of scarce institutional resources while satisfying the diverse operational, pedagogical, and personal needs of various stakeholders. The university exam timetabling problem (UETP), a prominent subset of this domain, is mathematically classified as NP-hard, indicating that no known algorithm can solve all possible instances in polynomial time.

The primary objective of UETP is to assign a discrete set of examinations to a limited set of timeslots and physical rooms. This assignment must strictly adhere to inviolable *hard constraints* — such as preventing overlapping exams for any enrolled student and respecting the physical seating capacities of rooms. Simultaneously, the system must navigate a multi-objective landscape of *soft constraints* to minimise

penalties associated with stakeholder preferences, consecutive examination fatigue, and workload distribution among faculty and staff.

Despite extensive research into genetic algorithms, simulated annealing, and hybrid metaheuristics for UETP, highly specific stakeholder conflicts frequently persist when theoretical models are applied to real-world environments. A critical gap in current modelling is the *dual-role conflict*. In modern universities, teaching assistants and Ph.D. candidates are often enrolled in graduate-level courses while simultaneously mandated to serve as invigilators for undergraduate examinations. Ensuring that their graduate exam schedules do not overlap with assigned invigilation duties requires specialised constraint formulation that bridges the traditionally separate sets of “students” and “instructors.”

This paper addresses this gap by introducing a comprehensive multi-objective CSOP model. Our framework introduces a novel multi-room splitting architecture enabling massive exams to span multiple physical venues simultaneously, and implements a strict min-max fairness proxy to prevent disproportionate assignment of duties to junior faculty.

The primary contributions of this paper are:

- 1) A formal mathematical constraint network modelling dual-role instructor conflicts as a hard constraint (H4).
- 2) A robust multi-room Boolean assignment matrix allowing large-enrollment exams to be split dynamically across multiple physical rooms.
- 3) A min-max workload fairness optimisation strategy providing a tighter guarantee against individual overwork than standard variance minimisation.
- 4) A comprehensive complexity analysis, four-instance Carter benchmark evaluation, and ablation study, including identification of a structural scalability boundary condition.

II. LITERATURE REVIEW

A. Classical Timetabling, Graph Colouring, and the Carter Benchmark

The modern examination timetabling literature traces its benchmark foundation to Carter, Laporte, and Lee [1], who

introduced the EXAMINE system and 13 real-world problem instances from Canadian and British universities — universally known as the Toronto or Carter benchmarks. These instances define the standard evaluation protocol: a proximity cost function that penalises conflicting student exams scheduled close together, with weights 16, 8, 4, 2, and 1 for timeslot gaps of 1–5 respectively, framing the problem as graph colouring over a student conflict graph. A key observation, later formalised by Ceschia et al. [4], is that the Toronto benchmark is deliberately *uncapacitated*: rooms are absent, invigilators are absent, and the sole hard constraint is conflict-freedom. This directly motivates our choice to model rooms and invigilators explicitly.

Burke and Newall [2] demonstrated that adaptive heuristic ordering — dynamically selecting graph colouring heuristics based on instance characteristics — consistently produces results at or above the state of the art across all 13 Carter instances. Qu et al. [3] subsequently provided the definitive survey of examination timetabling search methodologies over the preceding decade, establishing the authoritative comparative tables that subsequent work cites. Studies have further shown that the modularity of constraint networks strongly correlates with algorithm search time [9].

B. Metaheuristics in Educational Scheduling

To navigate the vast search spaces inherent to UETP, modern research has relied heavily on metaheuristics. Genetic algorithms (GA) have been extensively applied to multi-objective exam timetabling, balancing student convenience against resource penalties [10]. Ghaffar et al. [11] conducted a systematic review concluding that hybrid designs combining constructive heuristics with local search consistently outperform pure GA methods, while also noting the near-universal absence of fairness and invigilator modelling in reviewed formulations. Local search frameworks including Iterated Local Search (ILS) [19], Tabu Search, and Ant Colony Optimisation (ACO) combined with Lévy Flight [18] have been successfully adapted to escape local optima in complex timetabling datasets.

At the state-of-the-art end, Mandal, Kahar, and Kendall [5] proposed a partial graph heuristic with modified great deluge (PGH-mGD) achieving best-known or second-best proximity cost results on most Carter instances. Multi-neighbourhood local search with room-split balancing [17] is the nearest prior work to our anti-fragmentation term S5, though it operates as a heuristic without hard feasibility guarantees.

C. Constraint Programming and Exact Solvers

The Constraint Satisfaction Problem (CSP) provides a rigorous mathematical foundation. Alrawashdeh et al. [9] demonstrated that profiling-based CSP algorithms achieve feasible schedules with efficient classroom utilisation, validating constraint networks for real institutional data. Modern exact solvers, particularly those utilising SAT and Lazy Clause Generation (LCG) such as Google OR-Tools CP-SAT, have bridged the gap between theoretical constraint programming

and practical performance. Recent advancements explore the integration of Large Language Models (LLMs) and Graph Neural Networks (GNNs) with classical constraint reasoning to predict constraint propagation and solution validity [12], [20].

D. Dual-Role Instructor Conflicts and Workload Fairness

Existing literature treats the instructor pool and student body as mutually exclusive sets. Bazari et al. [6] modelled professor exam-authoring conflicts using MIP — a related but distinct problem — while Comlan and Allohoumbo [7] acknowledge dual-role doctoral student conflicts in the LMD framework but resolve them through manual rules rather than formal constraint encoding. To our knowledge, the present work is the *first* to encode dual-role conflict as a formal CP hard constraint (H4) within an exact solver. Gultekin et al. [8] include a workload balancing heuristic for midterm scheduling as a soft post-processing step; our min-max proxy (S2) is the first to integrate fairness as a primary weighted optimisation objective within an exact CP-SAT framework.

E. AI-Augmented Constraint Reasoning

Bonlarron et al. [12] demonstrated at IJCAI-25 that LLMs can be coupled with constraint propagation engines to translate natural-language problem descriptions into formal constraint clauses — directly relevant to real-world deployments where administrative constraints are expressed in natural language. GNNs are being explored as approximate CSP solvers [20], representing a promising warm-starting direction for large instances that exceed the current formulation’s memory capacity.

III. PROBLEM FORMULATION

The UETP is formulated as a CSOP. Feasibility is governed by inviolable hard constraints H1–H6; solution quality is measured by a weighted combination of soft constraints S1–S5. The problem instance is formally represented as a tuple $\mathcal{P} = (E, T, R, I)$, where:

- E is the set of examinations.
- T is the set of available timeslots.
- R is the set of rooms (physical and virtual).
- I is the set of instructors eligible for invigilation.

Each timeslot $t \in T$ is characterised by a *(day, period)* pair. Each room $r \in R$ has a finite capacity $cap(r)$, with one virtual room of capacity 100,000 serving as an unlimited-concurrency sink for online examinations. Each examination $e \in E$ is annotated with enrolled students $S(e)$, responsible lecturer $lec(e) \in I$, and required invigilator count $req(e)$.

A. Decision Variables

Timeslot Assignment X_e : an integer variable representing the timeslot assigned to examination e :

$$X_e \in \{0, 1, \dots, |T| - 1\} \quad \forall e \in E \quad (1)$$

Room Assignment Y_{er} : a Boolean matrix indicating if examination e utilises room r , enabling large-enrollment exams to span multiple classrooms simultaneously:

$$Y_{er} \in \{0, 1\} \quad \forall e \in E, \forall r \in R \quad (2)$$

Invigilator Assignment Z_{ei} : a Boolean variable indicating whether instructor i invigilates examination e :

$$Z_{ei} \in \{0, 1\} \quad \forall e \in E, \forall i \in I \quad (3)$$

B. Student Conflict Graph

A central data structure is the student conflict graph $G = (E, C)$, where an edge $(e_1, e_2) \in C$ exists if and only if $S(e_1) \cap S(e_2) \neq \emptyset$. Construction runs in $O(|E|^2 \cdot |\bar{S}|)$ time via set intersection, where $|\bar{S}|$ is the average class size. Algorithm 1 illustrates the procedure.

Algorithm 1 Conflict Graph Construction

Require: Set of exams E , student enrollments $S(e) \forall e \in E$

Ensure: Conflict matrix C where $C_{jk} = 1$ iff exams j, k share students

```

1: Initialise  $C_{jk} \leftarrow 0$  for all  $j, k \in \{1 \dots |E|\}$ 
2: for each pair  $(e_j, e_k) \in E \times E$  do
3:   if  $j < k$  then
4:      $Shared \leftarrow S(e_j) \cap S(e_k)$ 
5:     if  $|Shared| > 0$  then
6:        $C_{jk} \leftarrow 1$ 
7:     end if
8:   end if
9: end for
10: return  $C$ 

```

IV. MATHEMATICAL CONSTRAINTS METHODOLOGY

A. Inviolable Hard Constraints (H1–H6)

The following conditions represent operational impossibilities; any solution violating even one is rejected as inadmissible.

H1 — No Student Time Conflict. For every edge $(e_1, e_2) \in C$:

$$X_{e_1} \neq X_{e_2} \quad (4)$$

H2 — Room Capacity and Online Routing. The aggregate capacity of all assigned physical rooms must meet or exceed enrollment. Online exams are routed exclusively to the virtual room; overflow to the virtual room incurs a penalty of 5,000 units per exam:

$$\sum_{r \in R} Y_{er} \cdot cap(r) \geq |S(e)| \quad \forall e \in E \quad (5)$$

H3 — No Room Clash. Let $same_slot_{e_1 e_2}$ be a reified Boolean for $(X_{e_1} = X_{e_2})$:

$$same_slot_{e_1 e_2} \Rightarrow Y_{e_1 r} + Y_{e_2 r} \leq 1 \quad \forall r \in R_{\text{physical}} \quad (6)$$

H4 — Dual-Role Instructor Conflict (Novel Contribution). An instructor cannot invigilate an exam occurring in the same timeslot as an exam they are responsible for:

$$same_slot_{e_1 e_2} \Rightarrow Z_{e_2, lec(e_1)} = 0 \quad \forall e_1, e_2 \in E \quad (7)$$

H5 — No Double Invigilation. An instructor may invigilate at most one examination per timeslot:

$$same_slot_{e_1 e_2} \Rightarrow Z_{e_1 i} + Z_{e_2 i} \leq 1 \quad \forall i \in I \quad (8)$$

H6 — Minimum Invigilator Count. Each exam must be assigned exactly its required number of invigilators, where $req^*(e) = \min(req(e), |I|)$:

$$\sum_{i \in I} Z_{ei} = req^*(e) \quad \forall e \in E \quad (9)$$

B. Soft Constraints and Objective Function

The system minimises the weighted sum:

$$\min F = w_1 S_1 + w_2 S_2 + w_3 S_3 + w_4 S_4 + S_5 + 5000 \cdot |\text{overflow}| \quad (10)$$

with default weights $w_1=1, w_2=5, w_3=2, w_4=3$.

S1 — Time Preference. Penalises assigning instructors on declared leave days, implemented via `add_element` for variable-indexed array lookup and `add_min_equality` as a Boolean AND gate.

S2 — Workload Fairness (Min-Max Proxy). Let $load_i = \sum_e Z_{ei}$. Instead of quadratic variance (unsupported natively in CP-SAT), we adopt:

$$S_2 = \max_{i \in I} (load_i) - \min_{i \in I} (load_i) \quad (11)$$

This linear proxy provides a *stronger* fairness guarantee than variance minimisation: it directly prevents any single instructor from being disproportionately burdened.

S3 — Consecutive Invigilation. Penalises back-to-back assignments for the same instructor on the same day. Implemented via $O(|I| \times |\Gamma| \times |E|)$ nested reification, where Γ is the set of adjacent same-day timeslot pairs.

S4 — Student Day Gap. Penalises student exam pairs with $|day(e_1) - day(e_2)| \leq 1$, where $day(e) = \lfloor X_e / P \rfloor$ and P is the number of periods per day.

S5 — Anti-Fragmentation. Minimises $\sum_{e,r} Y_{er}$ for physical rooms, discouraging unnecessary room splitting when a single room suffices.

V. SYSTEM ARCHITECTURE AND IMPLEMENTATION

A. Data Parsing and Ingestion

A custom Python parser extracts exam, student, and instructor hierarchies from administrative spreadsheets. The parser employs fuzzy string matching via Levenshtein distance to reconcile disparate spellings of instructor names and course codes across departmental sheets, including Turkish character normalisation, slash-separated codes, and prefix-based patterns. A 94.4% match rate was achieved on the University X dataset; unmatched entries are attributable to cross-departmental courses absent from enrollment records.

B. CP-SAT Solver Integration

We utilise Google OR-Tools CP-SAT due to its advanced constraint propagation algorithms and native support for Large Neighbourhood Search (LNS). Hard constraints are encoded as mandatory clauses via `model.Add()`; soft constraints contribute penalty terms aggregated in `model.Minimize()`. The solver integrates AC-3 constraint propagation, Minimum

Remaining Values (MRV) branching heuristic, LNS for feasible solution improvement, and branch-and-bound for optimality proofs. All experiments used the CP-SAT default random seed (0) and OR-Tools version 9.x.

C. Asymptotic Complexity Analysis

Table I summarises the variable and constraint counts per model component. The dominant memory bottleneck is the H3 room-clash constraint at $O(|E|^2 \times |R|)$ — approximately 480,000 exclusion constraints for the University X instance and a projected 8.1 million for the 638-exam uta-s-92 instance, exceeding practical memory limits. The S3 consecutive-invigilation penalty contributes the largest auxiliary variable count via nested reification and is treated as optional for large instances.

TABLE I
ASYMPTOTIC COMPLEXITY OF MODEL COMPONENTS

Component	Variables	Constraints
Timeslot (X)	$O(E)$	—
Room (Y)	$O(E \times R)$	$O(E)$ capacity
Invigilator (Z)	$O(E \times I)$	$O(E)$ count (H6)
H1 (Conflict)	—	$O(C)$ pairwise
H3 (Room Clash)	$O(E ^2)$ reified	$O(E ^2 \times R)$
H4 & H5	Shared w/ H3	$O(E ^2 \times I)$
S1 (Preference)	$O(E \times I)$	$O(E \times I)$
S3 (Consecutive)	$O(I \times T \times E)$	$O(I \times T \times E)$

VI. EXPERIMENTAL SETUP AND RESULTS

The framework was evaluated across two environments: (A) four instances from the Carter/Toronto benchmark suite and (B) a real-world dataset from University X (Istanbul Okan University, Faculty of Engineering and Natural Sciences, Fall 2024–2025). All experiments used objective weights $w_1=1, w_2=5, w_3=2, w_4=3$, CP-SAT default seed (0), S3 disabled, S4 enabled, and a 600-second wall-clock limit. Carter instances used 40 synthetic instructors, 20 synthetic physical rooms, and 45 synthetic timeslots. Table II summarises instance characteristics.

TABLE II
DATASET CHARACTERISTICS AND CONFLICT METRICS

Parameter	hec-s-92	sta-f-83	yor-f-83	ear-f-83	Okan Univ.
Exams $ E $	80	138	180	189	131
Students $ S $	2,823	549	919	1,108	1,190
Conflict density	42.8%	19.3%	30.6%	27.3%	17.9%
Timeslots $ T $	45 ^a	45 ^a	45 ^a	45 ^a	62
Rooms $ R $	20 ^a	20 ^a	20 ^a	20 ^a	29
Instructors $ I $	40 ^a	40 ^a	40 ^a	40 ^a	36

^aSynthetically generated.

A. Carter Benchmark Results

The framework was tested on four Toronto benchmark instances of increasing scale. All four satisfied every hard constraint (H1–H6) with zero overflow. A fifth instance, uta-s-92 (638 exams, 21,329 students), was attempted but terminated by the operating system due to memory exhaustion before producing a solution. Complete results are given in Table III.

TABLE III
CARTER BENCHMARK RESULTS (S1+S2+S4, S3 DISABLED, 600 s)

Instance	Status	F	S1	S2 gap	S4	Time
hec-s-92 [†]	OPTIMAL	0	0	0 (9–9)	—	14.3 s
hec-s-92	FEASIBLE	2,292	0	2 (7–5)	734	600 s
sta-f-83	FEASIBLE	5,745	0	3 (6–3)	1,864	600 s
yor-f-83	FEASIBLE	5,660	1	2 (6–4)	1,823	600 s
ear-f-83	FEASIBLE	7,458	0	21 (22–1) [‡]	2,388	600 s
uta-s-92	OOM	—	—	—	—	killed

[†]S1+S2 only configuration; full S1+S2+S4 run yields $F=2,292$.

[‡]Structural saturation: all 45 timeslots and 20 rooms occupied.

hec-s-92 (80 exams). In the S1+S2-only configuration, the solver achieved a provably OPTIMAL solution in 14.3 s with a fairness gap of 0 — all 40 synthetic instructors received equal loads. In the full S1+S2+S4 configuration, objective $F=2,292$ with fairness gap of 2 (loads 5–7), zero preference violations, and only 35/45 timeslots used, confirming substantial scheduling slack.

sta-f-83 (138 exams). With 10.4 exams per student on average — the highest enrollment ratio in the suite — sta-f-83 creates dense conflicts despite moderate 19.3% conflict density. The solver achieved $F=5,745$ with fairness gap of 3 (loads 3–6) and zero preference violations.

yor-f-83 (180 exams). This instance saturated all 45 available timeslots and all 20 physical rooms, placing maximum structural pressure on the constraint model. Despite full saturation, the fairness gap was only 2 (loads 4–6) with just 1 preference violation — the only S1 violation across the benchmark suite, attributable to resource saturation leaving no timeslot flexibility.

ear-f-83 (189 exams) — Fairness Stress Case. The fairness gap collapsed to 21 (loads 1–22), dramatically worse than the other three instances. Three factors interact: simultaneous timeslot and room saturation (45/45, 20/20), conflict graph hotspots that force dense assignment to a small instructor subset, and S4’s penalty pool dominating the objective at this scale. This is not a solver failure but a *structural infeasibility*: no invigilator assignment policy can achieve fairness when all scheduling degrees of freedom are simultaneously exhausted. Administrators near these structural limits should expand the instructor pool or increase available timeslots.

uta-s-92 (638 exams) — Scalability Boundary. The $O(|E|^2 \times |R|)$ H3 constraint explosion (8.1 million exclusion constraints) exceeded practical memory limits. Column generation decomposition is required for instances beyond approximately 300 exams.

B. Positioning Relative to Prior Methods

The Carter proximity cost metric evaluates only student exam spacing — no rooms, no instructors, no fairness. Our CSOP model optimises a strictly richer problem formulation. Table IV positions this work against representative prior methods.

TABLE IV
COMPARISON WITH REPRESENTATIVE PRIOR METHODS

Method	Type	Prox. cost	Fairness	H4
Carter et al. (1996) [1]	Graph color.	~ 10.83	N/R	×
Burke & Newall (2004) [2]	Tabu Search	~ 10.10	N/R	×
Tung et al. (2021) [10]	GA	N/R	N/R	×
Mandal et al. (2020) [5]	Great Deluge	Competitive	N/R	×
This work	Exact CSOP	N/A*	0–3	✓

N/R = not reported. *Proximity cost not optimised; see discussion.

C. Real-World Application: University X Dataset

The model was applied to real Fall 2024–2025 exam data from University X: 131 examinations, 36 instructors (25 Ph.D. dual-role), 29 rooms (28 physical + 1 virtual), and 62 timeslots across 8 examination days. Three configurations were evaluated; results are given in Table V.

TABLE V
UNIVERSITY X RESULTS (131 EXAMS, 36 INSTRUCTORS)

Config	Status	S1	S2 gap	S4	Rooms	Ovfl.	Time
S3 off, 300 s	FEASIBLE	0	2 (6–4)	1,631	200	0	300 s
S3 off, 600 s	FEASIBLE	0	2 (6–4)	1,310	195	0	600 s
S3 on, 600 s	FEASIBLE	20	9 (10–1)	1,877	202	0	600 s

The best configuration (S3 off, 600 s) achieved: zero preference violations; a fairness gap of 2 with instructor loads bounded between 4 and 6 (mean 5.0, total 180 assignments); zero overflow; and effective multi-room splitting of a 446-student examination across 14 physical classrooms simultaneously. Increasing solver time from 300 s to 600 s improved S4 by 19.7% ($1,631 \rightarrow 1,310$), demonstrating continued solver convergence. Zero dual-role violations (H4) occurred across all runs.

D. Ablation Study: The Cost of S3

Table VI quantifies the tradeoff when S3 is enabled within the same 600-second budget.

Enabling S3 degraded every other objective simultaneously because it introduces approximately 236,000 auxiliary Boolean variables via nested reification, overwhelming the solver’s optimisation capacity within the time budget. S3 should remain disabled for instances exceeding ~ 130 exams unless solve-time budgets exceed approximately 30 minutes.

TABLE VI
ABLATION STUDY: S3 ENABLED VS. DISABLED (UNIVERSITY X, 600 s)

Metric	S3 off	S3 on	Change
S1 (pref. violations)	0	20	+20
S2 (fairness gap)	2	9	+7 (+350%)
S4 (day penalties)	1,310	1,877	+567 (+43%)
Objective F	4,135	5,898	+42.6%

Weights: $w_1=1, w_2=5, w_3=2, w_4=3$ (identical in both runs).

VII. DISCUSSION AND LIMITATIONS

Exact vs. heuristic methods. The CP-SAT solver proves optimality on hec-s-92 under the S1+S2-only configuration (14.3 s) — a guarantee no metaheuristic can provide. For larger instances, high-quality FEASIBLE solutions are returned within practical time limits.

The ear-f-83 boundary condition. The fairness collapse on ear-f-83 (gap 21 vs. 2–3 on other instances) is a structural finding: the min-max proxy works well when scheduling degrees of freedom are available but degrades under simultaneous timeslot and room saturation. Institutions near capacity should expand the instructor pool, increase timeslots, or raise w_2 relative to w_4 .

Proximity cost vs. multi-objective fairness. Comparing our model on proximity cost would require removing H4, H5, H6, and S2 — reducing it to standard graph colouring. The correct comparison is a richer constraint set vs. a simpler one, as Table IV reflects.

Scalability boundary. The uta-s-92 instance (638 exams) identifies the memory ceiling of the monolithic formulation. Column generation decomposition — separating timeslot assignment from room/invigilator routing — is the appropriate solution for instances beyond ~ 300 exams.

Limitations. Four limitations are acknowledged. (1) Invigilators are assigned to examinations as a whole rather than to specific physical rooms within a multi-room cluster. (2) The S3 consecutive-invigilation formulation creates a variable explosion bottleneck at scale. (3) All results beyond the hec-s-92 S1+S2 run are FEASIBLE; optimality gaps at termination are not yet reported. (4) The ear-f-83 fairness collapse under saturation has not been resolved algorithmically.

Reproducibility. All experiments used $w_1=1, w_2=5, w_3=2, w_4=3$, CP-SAT default random seed (0), OR-Tools v9.x, S3 disabled, S4 enabled, 600 s time limit. Carter instances (version 2) were sourced from the Nottingham benchmark repository [3]. University X data is institutional; solver code will be released upon acceptance.

VIII. CONCLUSION AND FUTURE WORK

This paper presented a CSOP framework for university examination timetabling that (i) explicitly resolves dual-role Ph.D. instructor conflicts as a hard constraint, (ii) integrates a multi-room Boolean assignment architecture, and (iii) optimises invigilator workload fairness via a min-max proxy.

Validation on four Carter benchmark instances and one real-world institutional dataset yielded four key findings: optimal fairness on hec-s-92; near-optimal fairness (gap 2–3) on sta-f-83 and yor-f-83; a structural fairness collapse under simultaneous saturation on ear-f-83 that constitutes a novel boundary condition finding; and a +42.6% objective degradation when S3 is enabled within a fixed time budget, establishing a hard operational tradeoff.

Four directions for future work are identified:

- 1) **S3 reformulation** using CP-SAT interval variables or `add_no_overlap` to reduce the $\sim 236K$ auxiliary variable overhead.
- 2) **Column generation** decomposing large instances (>300 exams) into timeslot-assignment and room/invigilator-routing subproblems.
- 3) **Warm starting** from existing manual schedules to reduce solve time by up to 70%.
- 4) **Invigilator-room assignment** within multi-room exam clusters for operational completeness.

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